L6b: Single Index Model Portfolio Allocation

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Objectives for Today

Today we put the single index model of L6a to work: assemble its inputs for a portfolio, solve L5b's allocation problems with them, and compare the result with the data-driven portfolios in sample and out of sample.

Today's concepts

- Construct and diagnose SIM portfolio inputs: the SIM mean vector and covariance from the archive, why the mean vector reproduces the sample means, and which residual covariances the SIM omits.
- Solve SIM allocation problems: long-only risky and risky and risk-free portfolios, the tangent portfolio from the ray of solutions, and what bounds, a borrowing restriction, or a higher borrowing rate change.
- Evaluate what the model changes: weights and covariance regret in sample, frozen allocations on the 2025 test year, and what one split can and cannot say.

Lectures and Examples

Lecture:
CHEME-5660-L6b-Lecture-SIM-Portfolio-RF-Fall-2026.ipynb

Example 1:
CHEME-5660-L6b-Example-SIM-MinVar-RA-Fall-2026.ipynb

Example 2:
CHEME-5660-L6b-Example-SIM-MinVar-RRFA-Fall-2026.ipynb

The first example assembles the SIM inputs for thirteen firms, checks two exact facts about them, traces the long-only frontier from each input set, compares weights and regret, and scores the portfolios on 2025 data. The second adds the risk-free asset, reads the tangent portfolio off the ray of solutions, and scores complete portfolios that lend, hold, or borrow.

Concept Review: The Single Index Model and Its Covariance

L6a: for asset $i \in \mathcal{P}$ on day t, with growth rate $g_{i,t}$ (inverse years), market growth rate $g_{M,t}$ (SPY; M names the index), intercept α_i, exposure β_i, and residual $\varepsilon_{i,t}$:

$$g_{i,t} = \alpha_i + \beta_i \, g_{M,t} + \varepsilon_{i,t}$$

with $\mathbb{E}[\varepsilon_{i,t}] = 0$, $\mathrm{Cov}(\varepsilon_{i,t}, g_{M,t}) = 0$, and $\mathrm{Cov}(\varepsilon_{i,t}, \varepsilon_{j,t}) = 0$ for $i \neq j$. Stack the assets $(\mathbf{g}, \alpha, \beta, \varepsilon)$, let $\bar{g}_M = \mathbb{E}[g_M]$, $\sigma^2_{g,M} = \mathrm{Var}(g_M)$, and $\mathbf{D}_g = \mathrm{diag}(\sigma^2_{g,\varepsilon,1}, \dots, \sigma^2_{g,\varepsilon,|\mathcal{P}|})$; under the three assumptions:

$$\boxed{\bar{\mathbf{g}} = \alpha + \beta \, \bar{g}_M, \qquad \Sigma_g = \sigma^2_{g,M} \, \beta\beta^\top + \mathbf{D}_g}$$

Rank one plus diagonal: $2|\mathcal{P}| + 1$ numbers for the covariance and $|\mathcal{P}| + 1$ for the mean, in place of $|\mathcal{P}|(|\mathcal{P}| + 1)/2$ covariances. Uncorrelated residuals is the assumption that makes it a covariance model.

Concept Review: Portfolio Risk and the Archive

For fixed weights $\mathbf{w}$, the linearized growth rate $g_p = \mathbf{w}^\top \mathbf{g}$ (L5b) has portfolio beta $\beta_p = \mathbf{w}^\top \beta$ and:

$$\mathrm{Var}(g_p) = \underbrace{\beta_p^2\, \sigma_{g,M}^2}_{\text{market risk}} + \underbrace{\mathbf{w}^\top \mathbf{D}_g \mathbf{w}}_{\text{residual risk}}$$

- Everything above is a population statement.
- The L6a estimation example fitted the model by least squares for every security in the 2014 to 2024 data and saved the estimates: that archive is where today's inputs come from.
- Derivation (one line): $g_i - \mathbb{E}[g_i] = \beta_i(g_M - \bar{g}_M) + \varepsilon_i$; multiply for i, j, take expectations; the assumptions kill the cross terms and, for $i \neq j$, the residual product.

Catch-up: ../L6a/CHEME-5660-L6a-Example-SVD-SIM-Estimation-Fall-2026.ipynb

SIM Portfolio Inputs from the Archive

The archive holds, per security fitted against SPY on the N training days, the least-squares estimates $\hat{\alpha}_i$, $\hat{\beta}_i$, and the residual variance $s^2_{g,\varepsilon,i} = \|\mathbf{r}_i\|_2^2/(N-2)$ ($\mathbf{r}_i$ the fitted residual series), and once the training-period sample mean g'_M and variance $s^2_{g,M}$ of the market. Substituting estimates for population quantities gives the SIM inputs for a portfolio:

$$\boxed{\hat{\mathbf{g}}_{\text{SIM}} = \hat{\alpha} + \hat{\beta}\, g'_M, \qquad \hat{\Sigma}_{g,\text{SIM}} = s^2_{g,M}\,\hat{\beta}\hat{\beta}^\top + \hat{\mathbf{D}}_g, \qquad \hat{\mathbf{D}}_g = \operatorname{diag}(s^2_{g,\varepsilon,1}, \ldots, s^2_{g,\varepsilon,|\mathcal{P}|})}$$

- Market moments from the training period: taking them from a later window mixes an in-sample fit with out-of-sample information; the archive stores the training values so it cannot happen by accident.
- Growth-rate units throughout: means in inverse years, variances in inverse years squared, no Δt (L6a).

Two Exact Facts about the SIM Inputs

How do they compare with L5b's data-driven inputs, the sample means $\hat{\mathbf{g}}$ and sample covariance $\hat{\Sigma}_g$ of the same firms over the same days?

The SIM mean vector is the sample-mean vector. A least-squares line with an intercept passes through the point of sample means (the residuals sum to zero), so $\hat{\alpha}_i + \hat{\beta}_i g'_M = g'_i$ and $\hat{\mathbf{g}}_{\text{SIM}} = \hat{\mathbf{g}}$. The two input sets agree on the reward.

The SIM covariance is the sample covariance with the residual covariances removed. Residuals are orthogonal to the ones and to the market series, so:

$$\hat{\Sigma}_{g,ij} = \hat{\beta}_i \hat{\beta}_j \, s^2_{g,M} + \widehat{\text{Cov}}(\mathbf{r}_i, \mathbf{r}_j)$$

so off the diagonal $(\hat{\Sigma}_g - \hat{\Sigma}_{g,\text{SIM}})_{ij} = \widehat{\text{Cov}}(\mathbf{r}_i, \mathbf{r}_j)$ exactly, the residual covariances L6a measured; on the diagonal only $N-1$ versus $N-2$ on the residual term, $-\|\mathbf{r}_i\|_2^2/[(N-1)(N-2)]$, a factor of about 1.0004.

Where the SIM Covariance Errs, and Positive Definiteness

The SIM covariance error is not a vague approximation: it is the residual covariance, pair by pair, with the sign structure L6a found.

- Same industry: positively correlated residuals (banks with banks by several tenths; automakers; semiconductor firms). The SIM understates their correlation.
- Technology firms with banks: negatively correlated residuals, two or three tenths. The SIM overstates their correlation.
- The SIM correlation matrix has one pattern, off the diagonal $\hat{\beta}_i\hat{\beta}_j s^2_{g,M}$ over the two total standard deviations, so no blocks; the data have blocks.

Positive definiteness: for any $\mathbf{x}$, $\mathbf{x}^\top \hat{\Sigma}_{g,\mathrm{SIM}}\mathbf{x} = s^2_{g,M}(\hat{\beta}^\top \mathbf{x})^2 + \sum_i s^2_{g,\varepsilon,i}x_i^2 \geq 0$; positive definite whenever every residual variance is positive (sufficient, not necessary). It is what the closed forms need; it says nothing about whether the matrix describes future co-movement.

Risky Asset Portfolios using Single Index Models

L5b's long-only problem with SIM inputs, for a target growth rate $g_\star$:

$$\boxed{\begin{aligned} &\min_{\mathbf{w}} \mathbf{w}^\top \hat{\Sigma}_{g,\mathrm{SIM}} \mathbf{w} = \underbrace{s_{g,M}^2 (\hat{\beta}^\top \mathbf{w})^2}_{\text{market risk}} + \underbrace{\textstyle\sum_i s_{g,\varepsilon,i}^2 w_i^2}_{\text{residual risk}} \\ &\text{s.t. } \hat{\mathbf{g}}_{\mathrm{SIM}}^\top \mathbf{w} \geq g_\star, \qquad \mathbf{1}^\top \mathbf{w} = 1, \qquad 0 \leq w_i \leq 1 \end{aligned}}$$

- Market term depends on the weights only through the portfolio beta; the residual term is a weighted sum of squares that diversification shrinks.
- The target is a floor: sweeping $g_\star$ from the GMV growth rate upward traces the GMV portfolio and the efficient branch, same convex quadratic program, same package.

Same Means, Different Covariance: Regret

The mean vectors agree, so every difference between the two frontiers is a residual covariance. For the same weights (ordered pairs):

$$\mathbf{w}^\top\hat{\Sigma}_{g,\text{SIM}}\mathbf{w} = \mathbf{w}^\top\hat{\Sigma}_g\mathbf{w} - \sum_{i\neq j} w_i w_j \,\widehat{\text{Cov}}(\mathbf{r}_i,\mathbf{r}_j) + \sum_i \frac{\|\mathbf{r}_i\|_2^2}{(N-1)(N-2)}\,w_i^2$$

The SIM understates variance where the held residuals correlate positively and overstates it where they correlate negatively. The honest comparison evaluates the SIM weights under the data covariance at the same target: the regret $\mathbf{w}_{\text{SIM}}(g_\star)^\top\hat{\Sigma}_g\,\mathbf{w}_{\text{SIM}}(g_\star) - \mathbf{w}(g_\star)^\top\hat{\Sigma}_g\,\mathbf{w}(g_\star) \geq 0$, because the data-optimal weights minimize exactly that quantity. Example 1: a variance regret of 0.01 to 1.2, a gap of a few thousandths to a tenth of a standard-deviation unit; nearly the same firms, weights up to seven points apart.

Example 1: CHEME-5660-L6b-Example-SIM-MinVar-RA-Fall-2026.ipynb

Risky and Risk-Free Asset Portfolios using Single Index Models

Add L5b's risk-free asset: growth rate g_f, zero variance over the matched horizon (a bill or a STRIP held to maturity; sold earlier it carries L2b's rate risk). Risk-free fraction w_f, risky weights $\mathbf{w}$, $w_f + \mathbf{1}^\top \mathbf{w} = 1$. The problem the package solves:

$$\boxed{\min_{\mathbf{w}} \ \mathbf{w}^\top \hat{\Sigma}_{g,\text{SIM}} \mathbf{w} \quad \text{s.t.} \quad \hat{\mathbf{g}}_{\text{SIM}}^\top \mathbf{w} + \underbrace{(1 - \mathbf{1}^\top \mathbf{w})}_{w_f} \, g_f \geq g_\star, \qquad 0 \leq w_i \leq 1}$$

- Only the risky weights carry variance and only they are bounded; w_f is not sign-constrained: $w_f < 0$ borrows at g_f, and the package allows it.
- Eliminating w_f: a constraint on the excess growth rate, $(\hat{\mathbf{g}}_{\text{SIM}} - g_f \mathbf{1})^\top \mathbf{w} \geq g_\star - g_f$.

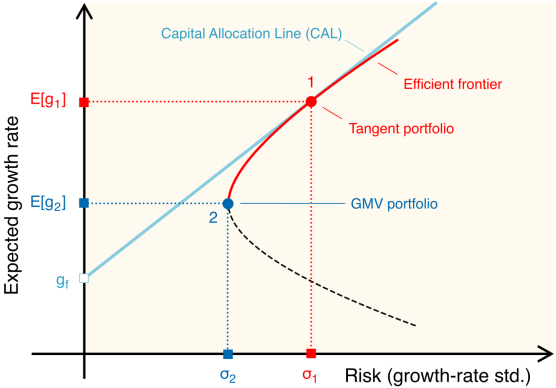
The Solutions Lie on a Ray

Scale the risky weights by a positive constant: excess growth scales by it, variance by its square, $w_i \geq 0$ unchanged. So, for feasible $g_\star > g_f$ ($\hat{\Sigma}_{g,\mathrm{SIM}} \succ 0$, so the minimizer is unique) and until an upper bound $w_i \leq 1$ binds, the minimizers are one risky direction scaled by the target. With $\mathbf{w}_\mathcal{T}$ the point on the ray with $\mathbf{1}^\top \mathbf{w} = 1$ ($w_f = 0$):

$$\boxed{\mathbf{w}(g_\star) = (1 - w_f)\,\mathbf{w}_\mathcal{T}, \qquad 1 - w_f = \frac{g_\star - g_f}{\mathbb{E}[g_\mathcal{T}] - g_f}}$$

- Below $\mathbb{E}[g_\mathcal{T}]$: lend ($0 < w_f < 1$). Above: borrow ($w_f < 0$). Valid until $(1 - w_f)\max_i w_{\mathcal{T},i} = 1$, then the boundary kinks.
- $\mathbf{w}_\mathcal{T} = \mathbf{w}/\mathbf{1}^\top \mathbf{w}$ from any solution on the ray (at $g_\star = g_f$ the solution is $\mathbf{0}$).
- Example 2 sweeps the target, checks every normalized solution is the same vector, and reads $\mathbf{w}_\mathcal{T}$ off the ray: the tangent portfolio of L5b.

The Picture



Portfolio 1 is the tangent portfolio $\mathcal{T}$, portfolio 2 the GMV portfolio; the CAL runs from $(0, g_f)$ through $\mathcal{T}$; the dashed lower branch is dominated.

The Capital Allocation Line and the Tangent Portfolio (Recall)

L5b: mixing a risky portfolio p ($\mathbb{E}[g_p]$, $\sigma_{g,p} > 0$) with fraction w_f risk-free gives $\mathbb{E}[g_c] = g_f + (1 - w_f)(\mathbb{E}[g_p] - g_f)$ and $\sigma_{g,c} = |1 - w_f|\,\sigma_{g,p}$; eliminating w_f (for $w_f \leq 1$) gives the capital allocation line $\mathbb{E}[g_c] = g_f + \mathrm{SR}_p\,\sigma_{g,c}$, a line from $(0, g_f)$ through p whose slope is the Sharpe ratio $\mathrm{SR}_p = (\mathbb{E}[g_p] - g_f)/\sigma_{g,p}$.

- A steeper line is better at every risk: hold the risky portfolio with the largest Sharpe ratio, the tangent portfolio $\mathcal{T}$; its CAL touches the frontier at $\mathcal{T}$.
- The ray is that CAL: least variance per unit of excess growth is the largest Sharpe ratio, so the ray's direction is the long-only maximum-Sharpe portfolio (Example 2 checks it against the frontier).
- Under SIM inputs, with $\hat{\beta}_{\mathcal{T}} = \hat{\beta}^{\top}\mathbf{w}_{\mathcal{T}}$:
$\mathrm{SR}_{\mathcal{T}} = (\hat{\alpha}^{\top}\mathbf{w}_{\mathcal{T}} + \hat{\beta}_{\mathcal{T}}\,g'_M - g_f)\Big/\sqrt{s^2_{g,M}\hat{\beta}^2_{\mathcal{T}} + \mathbf{w}^{\top}_{\mathcal{T}}\hat{\mathbf{D}}_g\mathbf{w}_{\mathcal{T}}}.$

Closed Form and Sharpe Conventions (Recall)

- Closed form (shorts allowed, $\hat{\Sigma}_{g,\mathrm{SIM}} \succ 0$, GMV portfolio earns more than g_f): $\mathbf{w}_{\mathcal{T}} \propto \hat{\Sigma}_{g,\mathrm{SIM}}^{-1}(\hat{\mathbf{g}}_{\mathrm{SIM}} - g_f \mathbf{1})$ normalized to sum to one; the rank-one-plus-diagonal structure gives the inverse in closed form (notes). Long-only: no closed form; the ray or the frontier sweep.
- Which Sharpe ratio: with $\sigma_{g,p}$ in the denominator it is the CAL slope in growth-rate units, a one-observation-interval quantity. Dividing by the volatility $\sqrt{\Delta t}\,\sigma_{g,p}$ (Δt the observation interval in years) gives the annualized value $\mathrm{SR}_p/\sqrt{\Delta t}$, larger by $\sqrt{252} \approx 15.9$. Same ranking; from here on the course quotes annualized.
- Example 2: about 1.3 annualized for both the SIM and the data-driven tangent portfolios on the training data; both hold the same three high-growth technology names, the SIM version less concentrated plus a small semiconductor position.

Separation, and What Breaks It

$\mathcal{T}$'s CAL dominates every other, so all mean-variance investors sharing estimates, g_f, horizon, and opportunity set hold one risky fund and differ only in w_f (Tobin, L5b): exact with one rate and no constraints; survives the long-only constraint on the risky weights. What breaks it:

- No borrowing ($w_f \geq 0$): truncates the straight CAL segment at $\mathcal{T}$; above it the efficient boundary generally returns to the risky frontier, so separation holds only piecewise. Not imposed by the package.
- Position bounds ($w_i \leq 1$, or a concentration limit $w_i \leq u$): kink the ray where they bind. Imposed in practice because unconstrained maximum-Sharpe portfolios put extreme weights on a few assets when the means are imprecise; a lower in-sample Sharpe, maybe more robust.
- Borrowing rate above the lending rate: kinks the line at $\mathcal{T}$.

Example 2: CHEME-5660-L6b-Example-SIM-MinVar-RRFA-Fall-2026.ipynb

What the SIM Inputs Change, and How to Check

- The SIM does not change the reward input; it replaces the sample covariance by a rank-one-plus-diagonal matrix that omits the residual covariances, with $2|\mathcal{P}| + 1$ numbers instead of $|\mathcal{P}|(|\mathcal{P}| + 1)/2$.
- Where most residual correlations are small (median 0.14 here, a few industry pairs 0.6 to 0.7) that costs little in sample: nearly the same assets, weights up to seven points apart, a standard-deviation gap of at most a tenth. It bites for two banks or two share classes of one firm.
- What it buys is not visible with thirteen firms: with hundreds of assets and a few years of daily data the sample covariance has more parameters than the data pin down and its inverse amplifies noise (L5b advanced); the SIM covariance is a few numbers per asset, always positive semidefinite, interpretable. Whether that helps out of sample is an empirical question about the universe.

Checking: Freeze, Test, Read for What It Is

- Freeze everything on the training window: universe, SIM parameters, market moments, g_f at the decision date, weights.
- Test the frozen weights with the buy-and-hold wealth rule on the test year, next to baselines: equal weights, the GMV portfolio, an index fund.
- Read it as one year, one draw, one universe. The paired SIM and data-driven portfolios earned nearly the same 2025 wealth: the input sets selected nearly the same portfolios, and one year cannot rank them (the gap is far smaller than the sampling error). Realized risk of the complete portfolios scaled almost exactly with $1 - w_f$ (buy-and-hold drifts, so not exactly); realized growth did not match the model, whose least certain input is the tangent portfolio's growth.
- Narrow beta intervals do not imply stable weights: propagating L6a's parameter uncertainty into the weights is the advanced notebook, and the question to ask before trading.

Optional Advanced Material

Advanced: advanced/uncertainty/
CHEME-5660-L6b-Advanced-SIM-Portfolio-Uncertainty-Fall-2026.ipynb

Optional, not a prerequisite for L7a. Bootstraps the SIM parameters of a six-firm universe two ways (empirical residuals and Gaussian innovations), keeps each asset's joint draw of intercept, beta, and residual scale, rebuilds the SIM covariance for every draw, and reads the distributions of a fixed unconstrained minimum-variance portfolio's growth-rate standard deviation, allocation distance, and variance regret.

Notebook: lectures/week-6/L6b/advanced, also linked from the lecture notebook.

Summary

Today: the SIM inputs for a portfolio, L5b's allocation problems with them, the CAL and separation as the package computes them, and the comparison in and out of sample.

Key takeaways

- The SIM changes only the covariance, and we know how: same sample means; sample covariance minus the off-diagonal residual covariances.
- One risky answer for every risk-free fraction: a free w_f makes the constraint one on excess growth, every target scales one direction by $1 - w_f$, and the fully invested point is the tangent portfolio; bounds, no borrowing, or a higher borrowing rate kink or truncate the ray.
- Regret in sample, frozen weights out of sample: SIM weights under the data covariance can only lose, and by little here; one test year checks weights without ranking input sets.

Next, L7a: a utility-based rebalancing engine, preferences recomputed daily.

Today: the same means, a covariance you can write the error of, and one risky answer for every risk-free fraction.